

```
# check the dataset  
# balanced or imbalanced
```

```
import pandas as pd
```

```
data = pd.read_csv("heart.csv")  
print(data)
```

```
print(data["output"].value_counts())
```

ML with Kamal Sir - 8369084928

create model

import lib

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
from pickle import dump
```

load the data

```
data = pd.read_csv("heart.csv")
```

check for null data

```
print(data.isnull().sum())
```

features and target

```
features = data.drop(["output"], axis="columns")
target = data["output"]
print(features)
print(target)
```

train and test

```
x_train, x_test, y_train, y_test = train_test_split(features.values, target)
```

model

```
model = LogisticRegression(max_iter=1000)
model.fit(x_train, y_train)
```

cr

```
y_pred = model.predict(x_test)
cr = classification_report(y_test, y_pred)
print(cr)
```

save the model

```
with open("heart.pkl", "wb") as f:
    dump(model, f)
    print("model saved")
```


use model

from pickle import load

load the model

make the prediction

age =

sex =

cp =

trtbps =

chol =

fbs =

restecg =

thalachh =

exng =

oldpeak =

slp =

caa =

thall =

d = [[age , sex, cp, trtbps, chol, fbs, restecg, thalachh, exng, oldpeak, slp, caa, thall]]

ans =

print()